

The Spectral Structure of One-Step Feature Learning from Spiked Data in Neural Networks

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Introduction

Neural networks learn representations from data, but which features do they learn, and which of these improve prediction?

We isolate this mechanism by studying the first gradient step; we find that even a low-rank weight update can generate nonlinear combinations of a few underlying directions. Building on low-rank gradient structure [1] and one-step feature-learning [2], we connect a **polynomial feature expansion**, **Gaussian equivalence**, and **limiting test error**.

- The structure of the data and the training scales determine which polynomial features emerge at the spectral scale of the feature bulk.
- Their predictive value depends on the new target information they capture beyond the untrained baseline and on how effectively this is exploited by the fitted predictor.

Model and training.

$$f_{W,a}(x) = \gamma_m a^\top \sigma(Wx), \quad W_1 = \rho W_0 - \eta \nabla_W \mathcal{R}_n(W_0).$$

d, m, n : input dimension, width, sample size; η, ρ : step size and weight-decay factor. Update W with fixed a ; fit a on an **independent sample**; evaluate on fresh test data.

Spiked Gaussian inputs.

$$x \sim \mathcal{N}(0, \Sigma_B + n^{2\nu} q q^\top), \quad 0 \leq \nu < \frac{1}{2}.$$

q is a unit spike direction; Σ_B is a possibly anisotropic, ill-conditioned bulk covariance.

Key assumptions.

- $n/d \rightarrow \psi_1 \in (0, 1)$, $m/d \rightarrow \psi_2 > 0$;
- spherical initialization; smooth activation with regular Hermite coefficients; concentrated training residuals.
- Risk results use a single-index teacher with bounded response second moment and fixed ridge penalty $\lambda > 0$.

A finite polynomial feature expansion

For fixed generalized Hermite truncation degree $L > 0$, the trained feature matrix satisfies

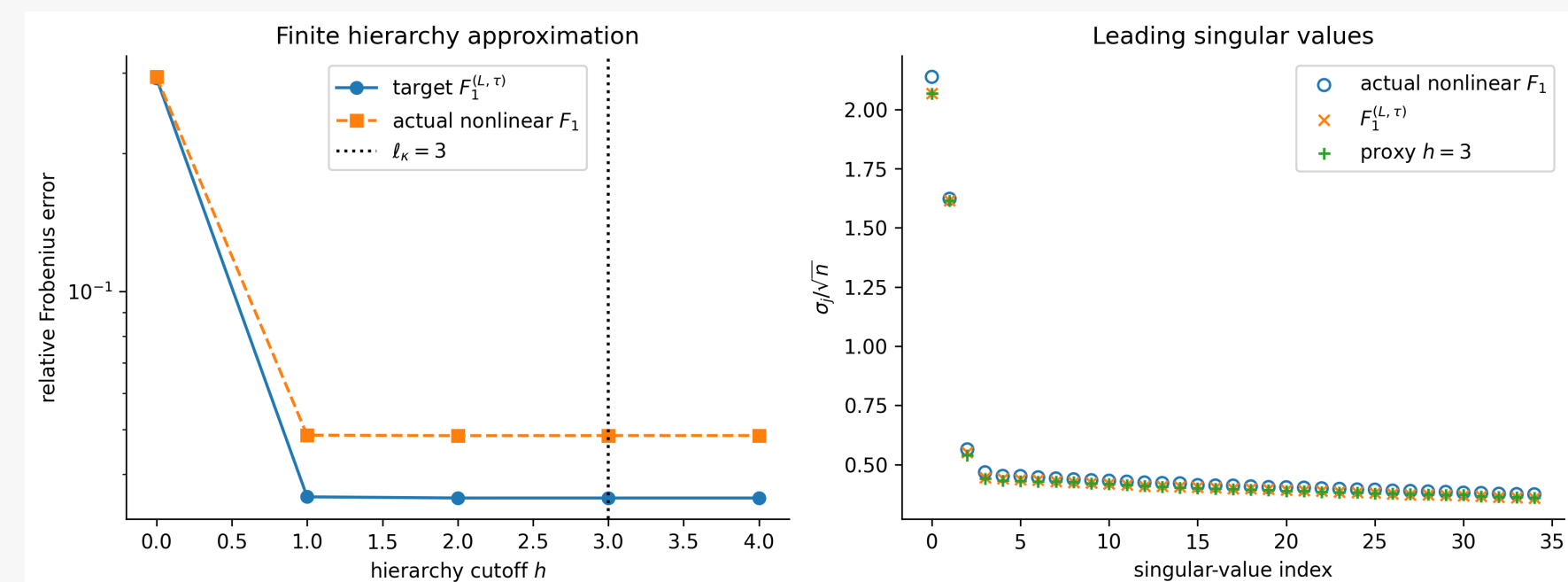
$$\underbrace{F_1}_{\text{trained features}} = \underbrace{F_0}_{\text{feature bulk}} + \underbrace{U_\ell V_\ell^\top}_{\text{polynomial hierarchy}} + E_n, \quad \frac{\|E_n\|_{\text{op}}}{\sqrt{n}} \xrightarrow{P} 0.$$

- F_0 is the weight-decayed, bulk-only random-feature bulk.
- U_ℓ 's columns contain the learned coordinates evaluated on the samples;
- V_ℓ gives their coefficients in each hidden unit's feature.

The hierarchy $U_\ell V_\ell^\top$ contains polynomials in **data-spike**, **data-residual-aligned**, and **residual-aligned** sources, including mixed terms and a constant correction; it satisfies

$$\text{rank}(U_\ell V_\ell^\top) \leq \binom{\ell+4}{4}.$$

Separation from the spectral bulk also depends on their geometry and the bulk spectrum.



A finite hierarchy approximates the truncated target and tracks the leading singular values of the actual nonlinear features. The full nonlinear matrix retains a truncation gap.

The feature-learning phase hierarchy

The degree counts the coordinate factors in a polynomial term: z is linear, z^2 or $z_1 z_2$ quadratic, and z^3 cubic. Which degrees can contribute at the feature bulk's spectral scale depends on the exponents of **network normalization** (γ_{sc}), **step size** (δ), and **data-spike size** (ν).

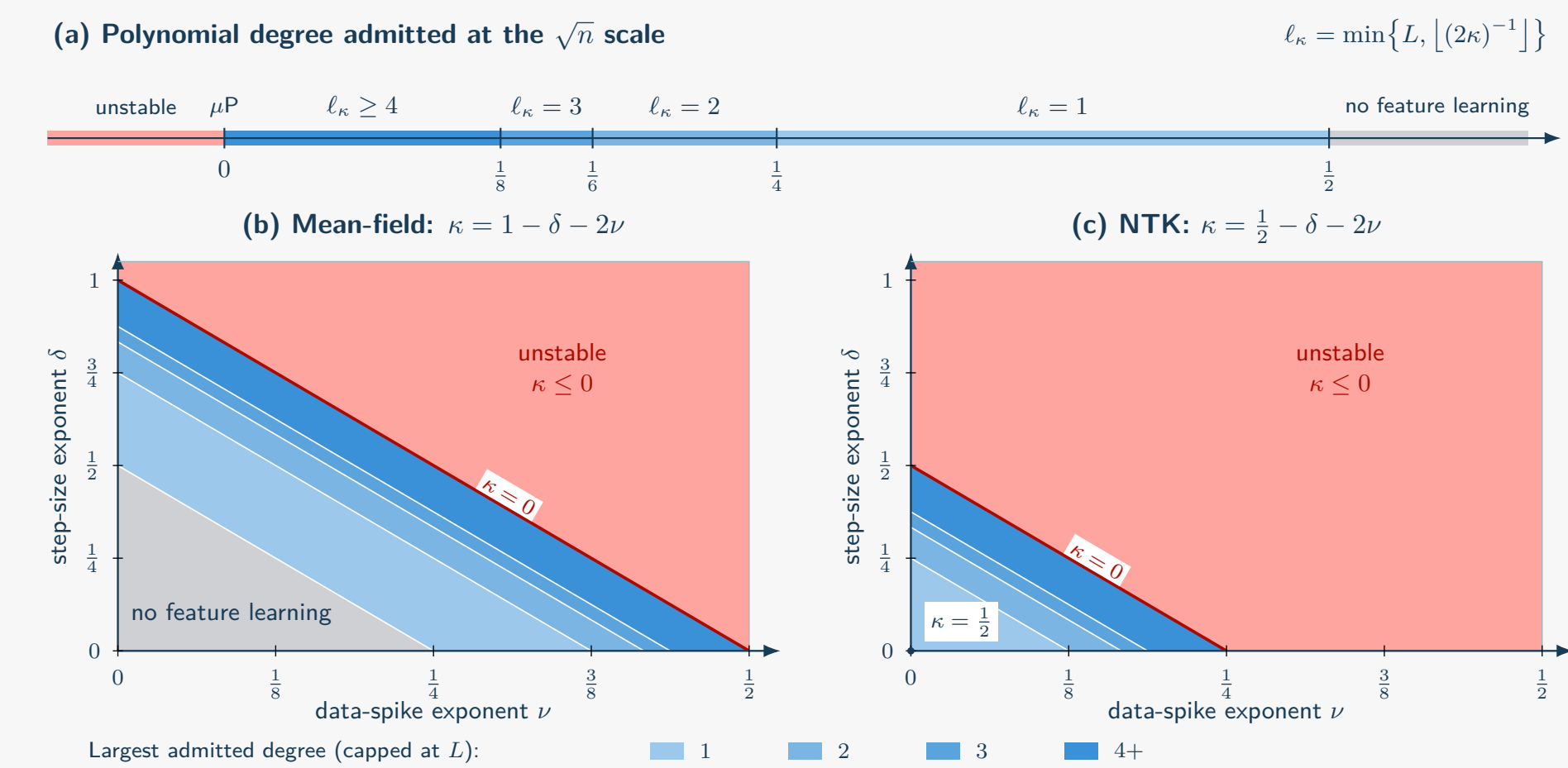
$$\gamma_m = m^{-1/2 - \gamma_{\text{sc}}}, \quad \eta \asymp m^\delta,$$

Neural tangent kernel (NTK) scaling is $\gamma_{\text{sc}} = 0$; mean field (MF) scaling is $\gamma_{\text{sc}} = 1/2$.

We identify a **threshold parameter** that governs feature learning:

$$\kappa = \frac{1}{2} + \gamma_{\text{sc}} - \delta - 2\nu.$$

Larger steps or stronger spikes admit higher degrees.



In our setting, $\kappa = 0$ is equivalent to μP parameterization [3]. For $\kappa > 1/2$, the perturbation is negligible.

Gaussian equivalence

A (corrected) **Gaussian surrogate** replaces the nonlinear feature bulk in admissible data-adaptive probes, extending random-feature universality [4] to the anisotropic setting.

$$\tilde{F}_0 = \underbrace{F_{\text{linear}} + Z_{\text{matched}}}_{\text{linear features} + \text{Gaussian noise}} + \underbrace{B_0 + B_2}_{\text{finite-rank corrections}}.$$

B_0 retains the degree-zero bulk direction; B_2 corrects the degree-two covariance. They can affect prediction even when invisible in the limiting spectral distribution.

For a ridge resolvent $R = (FF^\top/n + \lambda I)^{-1}$ and any fixed admissible probe family,

$$\frac{1}{n} u^\top (R_0 - \bar{R}_0) v \xrightarrow{P} 0, \quad \frac{1}{n} \text{tr}(A(R_0 - \bar{R}_0)) \xrightarrow{P} 0.$$

u, v are admissible probe directions with norm $O_P(\sqrt{n})$; A has bounded operator norm. Marked probes retain dependence on the same initialization that generates the learned coordinates.

Bridge to the test error.

Focusing on the feature bulk allows us to circumvent the issue of Gaussian equivalence failing for large step sizes; we treat the polynomial hierarchy term separately via the Woodbury identity.

References

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Limiting test error

At a cutoff $\ell \leq L$, the test-MSE difference satisfies

$$\mathcal{L}_{\text{te}}(F_0) - \mathcal{L}_{\text{te}}(F_1) \xrightarrow{P} \Lambda_\ell,$$

under convergence and stability of the corrected Gaussian surrogate. Thus Λ_ℓ is the limiting improvement over the reference.

Specifying Λ_ℓ .

In **saturation** (vanishing effective penalty on learned coordinates), Λ_ℓ is the **best prediction gain beyond the baseline minus a mismatch penalty**. This requires convergent residual moments, nondegenerate training coordinates, and vanishing test-prediction correction from the shared readout weights.

One-coordinate formula.

Subtract baseline ridge fits to obtain the test residual e and coordinate z . When trained and test-optimal coefficients agree,

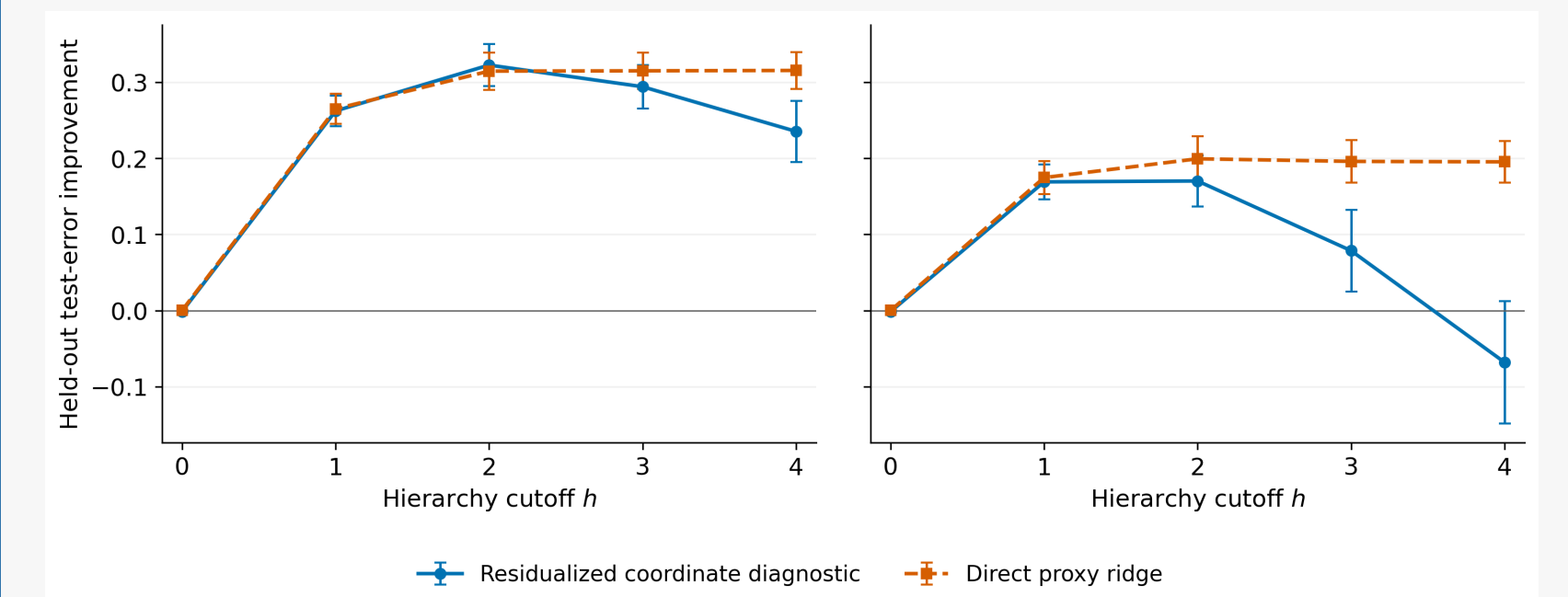
$$\Lambda_\ell = \frac{\mathbb{E}_{\text{te}}[ez]^2}{\mathbb{E}_{\text{te}}[z^2]}, \quad \mathbb{E}_{\text{te}}[z^2] > 0.$$

Expectations denote limiting conditional test moments.

Isotropic MF case.

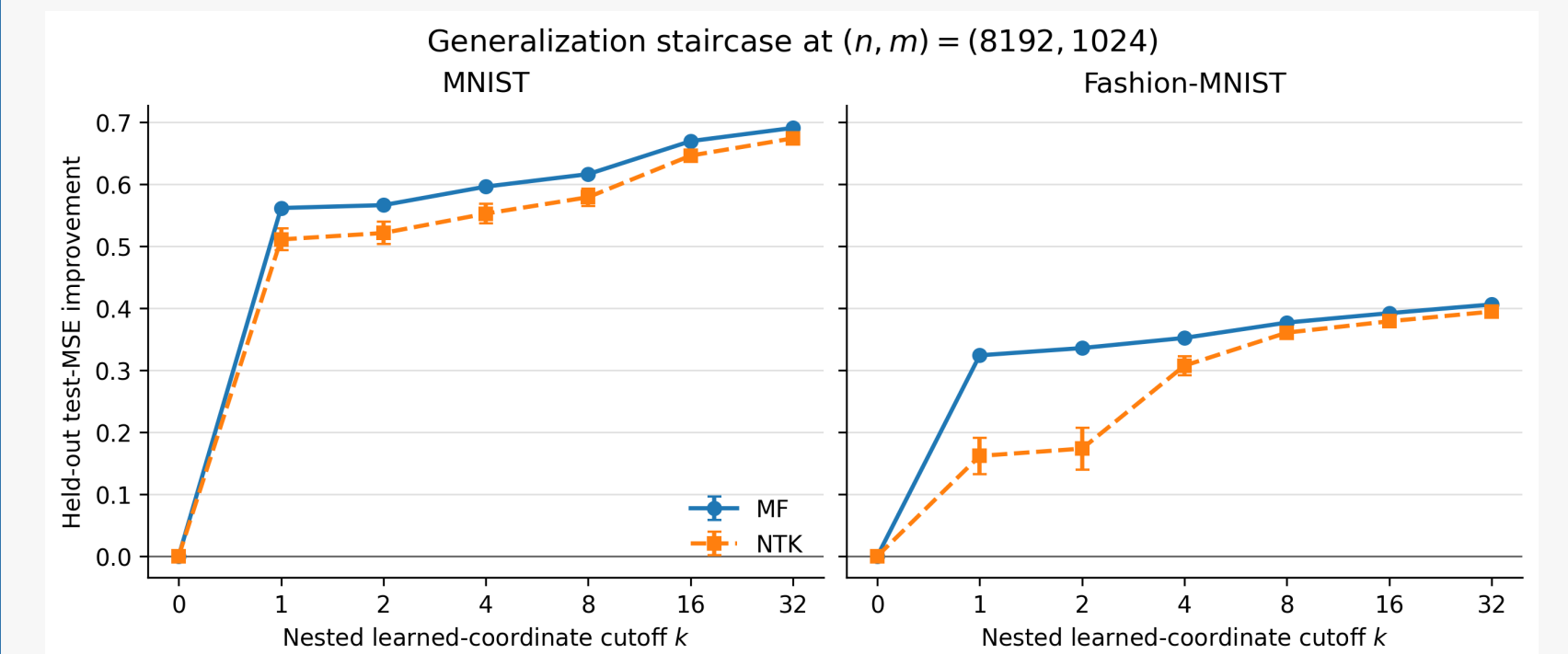
With no spike and a single active degree-one coordinate, Λ_ℓ can be reduced to a scalar formula under additional assumptions ensuring coefficient agreement [2].

Spectral visibility vs. predictive value



Network-generated coordinates; quadratic teacher (left), quartic teacher (right). Direct proxy ridge (orange) and residualized-coordinate diagnostic (blue) need not improve as the cutoff grows. Higher admitted degree alone does not ensure lower test error. Mean $\pm$ SD over 10 runs.

Learned-coordinate staircases on real data



Residualized learned singular coordinates give empirical staircases under MF and NTK scaling, with larger MF gains. Mean $\pm$ SD over 10 paired runs. Both scalings show monotone empirical staircases; MF has the larger gain, while the gap narrows with the coordinate cutoff.